

Validity Analysis: MMLU

Target context: Non-Arab Tourists and Expats in Eight Arab Countries

Overall risk: **HIGH**

Deployment Context

Use case: Support foreigners visiting Morocco, Egypt, Jordan, Palestine, Lebanon, UAE, Kuwait, or KSA in answering their general questions

Target population: Non-Arab tourists or expats visiting Morocco, Egypt, Jordan, Palestine, Lebanon, UAE, Kuwait, or KSA and have general questions about the Arabic language, history or geography of these countries

Dimension Scores

Dimension	Score	Rating	Confidence
Input Ontology $\triangle$	1	Serious concern	high
Input Content $\triangle$	1	Serious concern	high
Input Form $\checkmark$	3	Moderate gaps	high
Output Ontology $\triangle$	1	Serious concern	high
Output Content $\triangle$	1	Serious concern	high
Output Form $\checkmark$	2	Significant gaps	high
Average	1.5		

$\triangle$ = highest concern $\checkmark$ = strongest dimension

Dimension Key

Abbr.	Dimension	Definition
IO	Input Ontology	Whether the benchmark's test case categories cover the query types expected in deployment.
IC	Input Content	Whether individual datapoint content is culturally and contextually appropriate for the target region.
IF	Input Form	Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.
OO	Output Ontology	Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.
OC	Output Content	Whether ground-truth labels align with the judgments of regional stakeholders.
OF	Output Form	Whether the expected output modality matches regional deployment needs and accessibility.

Overall Summary

MMLU is fundamentally misaligned with this deployment across the four HIGH-priority dimensions (IO, IC, OO, OC). Documentation, web evidence, and dataset analysis converge on a clear finding: MMLU's 57 subjects are anchored in US academic curricula with no dedicated Arab-region content; ground-truth labels reflect US-academic and (in moral_scenarios) explicitly US-2020 framings; and the single-answer MCQ schema is structurally incapable of representing the multi-perspective output behavior required for contested Arab historical and political topics. The two MODERATE-priority dimensions (IF, OF) match deployment modality at the surface (English text) but cannot evaluate the Arabic-learning sub-cohort or open-ended explanatory output. Universal STEM and formal-reasoning content remains usable as a transferable competence baseline, but

the core deployment domain — Arab history, regional geography, Arabic language basics, and Islamic religious knowledge — is essentially uncovered.

Practical Guidance

What This Benchmark Measures

MMLU can usefully measure transferable academic competence in subjects with no cultural dependence — STEM, formal logic, mathematics — and can serve as a baseline for English-language reading comprehension and MCQ reasoning. It cannot measure the deployment's core constructs: Arab history knowledge, Middle Eastern geography, Arabic language basics, Islamic religious knowledge, or the ability to acknowledge multiple perspectives on contested topics.

Construct Depth

Construct depth is shallow for the deployment's purpose. The benchmark probes broad academic recall via MCQ but cannot surface (a) regional factual accuracy on Arab content, (b) framing perspective on contested events, (c) calibrated hedging on politically sensitive topics, or (d) Arabic-language competence. Even the calibration evaluation [Q64–Q67], while informative about confidence-accuracy gaps, cannot diagnose region-specific overconfidence patterns documented in CAMEL and MENAValues research [WEB-11, WEB-15].

What Else You Need

Substantial supplementation is required. For IO and IC: ArabicMMLU [WEB-2] (40 tasks, 8 countries, 14,575 MSA questions) and ILMAAM [WEB-5] (culturally refined with 11-expert annotation). For dialect/IF coverage of the Arabic-learning cohort: DialectalArabicMMLU [WEB-1] (Moroccan, Egyptian, Emirati, Saudi). For OC value alignment across all eight countries: MENAValues [WEB-11]. For Islamic-cultural content: PalmX 2025 [WEB-13] and IslamicMMLU [WEB-14]. None of these supplements solve the OO contested-topic schema gap, which is a confirmed ecosystem-wide gap (FV5) and would require new evaluation methodology (multi-perspective scoring, expert adjudication of contested labels).

ID	Evidence
Q64	"We should not trust a model's prediction unless the model is calibrated, meaning that its confidence is a good estimate of the actual probability the prediction is correct." (p.7)
Q65	"We evaluate the calibration of GPT-3 by testing how well its average confidence estimates its actual accuracy for each subject." (p.7)
Q66	"Its confidence is only weakly related to its actual accuracy in the zero-shot setting, with the difference between its accuracy and confidence reaching up to 24% for some subjects." (p.7)
Q67	"Another calibration measure is the Root Mean Squared (RMS) calibration error (Hendrycks et al., 2019a; Kumar et al., 2019)." (p.7)
WEB-1	DialectalArabicMMLU covers five Arabic dialects (Syrian, Egyptian, Emirati, Saudi, Moroccan) with substantial cross-dialect performance variation (https://arxiv.org/abs/2510.27543)
WEB-2	ArabicMMLU is the closest available substitute for the Arabic-language secondary cohort (14,575 MSA questions) (https://aclanthology.org/2024.findings-acl.334/)
WEB-5	ILMAAM (2025) identified significant cultural misalignments in religion and morality content; 11 expert annotators reviewed for cultural appropriateness, bias, religious sensitivity (https://aclanthology.org/2025.loreslm-1.29/)
WEB-11	MENAValues evaluates LLMs across neutral, personalized, and observer framings — illustrating richer output-form evaluation that MMLU's MCQ accuracy cannot match (https://arxiv.org/html/2510.13154)
WEB-13	https://arxiv.org/html/2509.02550
WEB-14	https://arxiv.org/html/2603.23750
WEB-15	Naous et al. 2024 (CAMEL) empirically demonstrate LLM Western cultural bias in Arabic prompts (https://venturebeat.com/ai/large-language-models-exhibit-significant-western-cultural-bias-study-finds)

Input Ontology (IO)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: MMLU | Context: Non-Arab Tourists and Expats in Eight Arab Countries

Input Ontology definition: Whether the benchmark’s test case categories cover the query types expected in deployment.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

MMLU's 57-subject taxonomy is anchored in US academic curricula with no dedicated Arab history, Middle Eastern geography, Arabic language, or Islamic jurisprudence subjects. Dataset analysis confirms that even subjects nominally relevant to the deployment (high_school_world_history, high_school_geography, global_facts, world_religions) contain little to no Arab-region content in sampled examples. Given IO is rated HIGH priority and the deployment’s core domain is precisely what MMLU omits, this is a fundamental construct underrepresentation.

Strengths

- Universal STEM and formal reasoning subjects (mathematics, physics, chemistry, formal logic, computer science) test culturally neutral content valid for any region, including Arab contexts (Strength 1; D1–D43 passim)
- World religions subject nominally covers multiple faith traditions, providing at least a structural slot where Islamic content could appear [Q171]
- History subject claims to span 'a wide range of time periods and geographical locations' [Q41], even if dataset evidence shows minimal Arab-region instantiation

ID	Evidence
Q41	"Finally, our history questions cover a wide range of time periods and geographical locations, including prehistory and other advanced subjects." (p.4)
Q171	"Figure 70: A World Religions example." (p.27)
D1	The right of American citizens to petition the government for a redress of grievances is protected under the First Amendment — US constitutional law — exclusively US institutional framing
D2	In the majority of cases, federal programs are implemented by state and local governments, by means of federal funding — US federal governance structure
D3	Both the War Powers Act of 1974 and the Budget and Impoundment Control Act of 1974 represent efforts by Congress to limit the powers of the president — US-specific legislative history
D4	John C. Calhoun, 'South Carolina Exposition and Protest,' 1828 — US political history — no Arab-region analog
D5	Tonight, the daughter of an immigrant from Italy has been chosen to run for (vice) president... Geraldine Ferraro, Vice Presidential Nomination Acceptance Address, July 19, 1984 — US domestic politics 1984
D6	A truck driver was assigned to drive a huge tractor-trailer loaded with logs... the girl was struck by the tractor-trailer and was seriously injured — US tort law scenario
D7	A patient filed a medical malpractice action against a hospital in federal court — US federal court procedure
D8	A buyer filed a lawsuit against a seller based on a written contract allegedly executed at the time of the sale of the seller's hot dog stand — US contract law
D9	Why is NSC 68 seen as a turning point in US Cold War Foreign policy? ... It indicated a shift towards military containment — US Cold War policy — framed entirely from US perspective
D10	In American government, the power to declare war rests with Congress. — US constitutional structure
D11	Revelations that the NSA was monitoring the communications of American citizens without obtaining warrants — US domestic surveillance scandal

D12	For which of these two scenarios does the main character do something clearly morally wrong, according to ordinary moral standards in the US as of 2020? — Explicitly frames moral judgment via US 2020 standards
D13	according to ordinary moral standards in the US as of 2020? Scenario 1: When I went through the road block I gave the policeman my brothers ID. — US-anchored moral framing
D14	according to ordinary moral standards in the US as of 2020? Scenario 2: I was due to start work soon, so I went to a baseball game. — US cultural reference (baseball) embedded in moral reasoning
D15	British journalist William Howard Russell, My Indian Mutiny Diary, 1860... 'the deed was done by a subject race — by black men who dared to shed the blood of their masters' — Colonial-era British framing of Indian revolt
D16	Excerpt from the inscription of Kerdīr at Naqsh-e-Rustam, Persia, late third century C.E. ... Jews, Buddhists, Hindus, Nasoreans, Christians, Maktak, and Manichaeans in the empire were smitten — Near Eastern (Persian) source — the only example touching the broader Middle East region
D17	Rome claimed that their bishop (pope) was the direct successor of which leader? ... Peter — Christian ecclesiastical history — no Islamic content in sampled examples
D18	In Jainism, what is the cycle that one must liberate oneself from? ... Samsara — Jainism content
D19	Which are the synoptic Gospels? ... Matthew, Mark, Luke — Christian Biblical content
D20	Which Jaina group only allows women to lead a life of semi-renunciation? ... Digambara — Jainism content — no Islamic content in sample
D21	What is the most important prayer in Judaism? ... The Shema — Jewish religious content
D22	About what percentage of survey respondents from China report having paid a bribe in the last year to access public services ... as of 2017? — China-focused global facts
D23	As of 2017, what fraction of the population in India used the internet in the past three months? ... 26% — India-focused global facts
D24	As of 2019, which of the following had the lowest life expectancy? ... Mexico — Global health — Latin American and Asian countries featured; no Arab countries
D25	As of 2018, about what percentage of Indians say that having an increasing number of people of many different races ... makes it a better place to live? — India-focused survey data
D26	At the time of independence, there were already hundreds of thousands of university graduates in ____, but hardly any at all in _____. ... India, Congo — Development economics — Africa/Asia framing, no Arab countries
D27	What is the most rapidly growing religion in the United States today? ... Islam — US-context geography question; Islam mentioned but in US demographic context
D28	Which zone contains low-income slums, ethnic ghettos, and general deterioration in Burgess's concentric zone model of urban form? — US urban sociology model (Burgess)
D29	Leeds Woolen Workers Petition, 1786 ... The sentiments expressed by the Leeds woolen workers illustrate which of the following historical trends? — British industrial history
D30	Voltaire, Letters on the English Nation, 1733 ... of all religions, the Christian ought doubtless to inspire the most tolerance — European Enlightenment philosophy
D31	civil society is far less developed than in, for instance, _____ ... Russia, China, Britain — Western-centric civil society framing; Arab world absent from options
D32	The age at which one can receive full benefits from social security ... Has been increasing — US social security system
D33	Which of the following groups of Americans have the lowest rate of alcohol abuse? ... Asian-American — US demographic framing
D34	Which of the following is the most common method of birth control for married couples in the U.S. ... Sterilization — US-specific reproductive health statistics
D35	From 1988 to 1990 among heterosexuals in the US, the number of unmarried adults aged 20 to 45 who report having multiple partners has: declined for both men and women — US sexual behavior survey data
D36	Archaeological evidence indicates that cattle were first domesticated where and how long ago? ... in the Middle East, about 10,500 years ago — Middle East referenced as origin of cattle domestication — one of very few regional mentions
D37	Kwame Nkrumah, Neo-Colonialism, 1965 ... 'A State in the grip of neo-colonialism is not master of its own destiny.' — African post-colonial context — non-Western source, though not Arab
D38	Which of the following statements best describes the postmodernist project? ... It rejects the Kantian concern with individual rights, equality, and justice. — Western jurisprudential theory; no Islamic jurisprudence
D39	What is the meaning of cultural relativism? ... Cultural relativism posits that local culture should validate the existence and practice of all human rights — International law — potentially relevant to Arab regional norms

D40	Jim watched a liquor store furtively for some time, planning to hold it up ... On a charge of burglary, Jim's best defense would be that the liquor store was open to the public. — US criminal law scenario
D41	As reported on CNN, in a May 1999 national poll 43% of high school students expressed fear about going to school — US-sourced statistics example
D42	Best described as intensification of worldwide social relations and increasing interdependence, globalization is the result of the compression of space and time — Abstract IR theory — not regionally specific
D43	With regard to minority and nonminority clients, psychotherapy is ... equally effective — US psychology research context

Checklist

Checklist item definitions:

ID	Assessment Question
IO-1	Identify test case categories required for regional deployment
IO-2	Check if taxonomy omits regionally relevant categories
IO-3	Check if taxonomy includes irrelevant categories
IO-4	Document category gaps harming content validity

Checklist responses:

IO-1: Required categories per the deployment YAML and elicitation include Arab history (Islamic Golden Age, Ottoman period, modern state formation), regional geography of eight countries, Arabic language basics, Islamic religious knowledge, and pan-Arab cultural background.

IO-2: Yes — MMLU has no dedicated subject for any of these categories. The 57 subjects include `high_school_us_history`, `high_school_european_history`, `us_foreign_policy`, `professional_law` (US bar exam), but no Arab/MENA equivalents [Q1, Q4, Q37, Q43]; dataset confirms exhaustively (D1, D3, D4, D9–D11).

IO-3: Yes — multiple subjects (`high_school_us_history`, `high_school_government_and_politics`, `us_foreign_policy`, `professional_law`, `human_aging`, `human_sexuality`) are US-institution-specific and contribute construct-irrelevant variance for an Arab-region deployment (D1–D11, D32–D35).

IO-4: Confirmed structural gap: zero dedicated Arab-region subjects out of 57; nominally relevant subjects (`global_facts`, `world_religions`, `high_school_geography`) contain little or no Arab-region content in 5–6 sampled examples each (D22–D28). ArabicMMLU and ILMAAM exist as supplements but are not part of MMLU [WEB-2, WEB-5].

ID	Evidence
Q1	“We propose a new test to measure a text model's multitask accuracy. The test covers 57 tasks including elementary mathematics, US history, computer science, law, and more.” (p.1)
Q4	“The benchmark covers 57 subjects across STEM, the humanities, the social sciences, and more. It ranges in difficulty from an elementary level to an advanced professional level, and it tests both world knowledge and problem solving ability.” (p.1)
Q37	“Branches of the humanities include law, philosophy, history, and so on (Appendix B).” (p.4)
Q43	“Subject areas include economics, sociology, politics, geography, psychology, and so on.” (p.4)
Q169	“Figure 68: A US Foreign Policy example.” (p.26)
D1	The right of American citizens to petition the government for a redress of grievances is protected under the First Amendment — US constitutional law — exclusively US institutional framing
D2	In the majority of cases, federal programs are implemented by state and local governments, by means of federal funding — US federal governance structure
D3	Both the War Powers Act of 1974 and the Budget and Impoundment Control Act of 1974 represent efforts by Congress to limit the powers of the president — US-specific legislative history
D4	John C. Calhoun, 'South Carolina Exposition and Protest,' 1828 — US political history — no Arab-region analog
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D32	The age at which one can receive full benefits from social security ... Has been increasing — US social security system
D33	Which of the following groups of Americans have the lowest rate of alcohol abuse? ... Asian-American — US demographic framing
D34	Which of the following is the most common method of birth control for married couples in the U.S. ... Sterilization — US-specific reproductive health statistics
D35	From 1988 to 1990 among heterosexuals in the US, the number of unmarried adults aged 20 to 45 who report having multiple partners has: declined for both men and women — US sexual behavior survey data
WEB-2	ArabicMMLU is the closest available substitute for the Arabic-language secondary cohort (14,575 MSA questions) (https://aclanthology.org/2024.findings-acl.334/)
WEB-5	ILMAAM (2025) identified significant cultural misalignments in religion and morality content; 11 expert annotators reviewed for cultural appropriateness, bias, religious sensitivity (https://aclanthology.org/2025.loreslm-1.29/)
WEB-12	Survey of Arabic benchmarks notes gaps in coverage of North Africa, Levant, Iraq even within the Arabic NLP ecosystem (https://arxiv.org/html/2510.13430v1)

Information Gaps

- No published quantitative audit of original English MMLU for Arab-region content proportion across all 15,908 items (FV1)
- Whether the full world_religions set (171 questions) contains Islamic content beyond the sample

Requires Expert Verification

- Full-set count of Arab-region or Islamic-content questions across world_history, world_religions, global_facts, and miscellaneous subjects

Remediation

Gap 1: No dedicated Arab history, Middle Eastern geography, Arabic language, or Islamic content subjects in MMLU's 57-subject taxonomy.

Recommendation: Replace MMLU as the primary Arab-region knowledge evaluation with ArabicMMLU [WEB-2] (40 tasks across 8 Arab countries) and PalmX 2025 [WEB-13] (Arabic and Islamic culture). Retain MMLU only for transferable

STEM/formal-reasoning subjects.

ID	Evidence
WEB-2	ArabicMMLU is the closest available substitute for the Arabic-language secondary cohort (14,575 MSA questions) (https://aclanthology.org/2024.findings-acl.334/)
WEB-13	https://arxiv.org/html/2509.02550

Input Content (IC)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: MMLU | Context: Non-Arab Tourists and Expats in Eight Arab Countries

Input Content definition: Whether individual datapoint content is culturally and contextually appropriate for the target region.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

All identified MMLU source materials are Anglophone Western institutions (GRE, USMLE, US undergraduate courses, Oxford University Press, Harvard Law Library) [\[Q23–Q25, Q83, Q85\]](#). No Arab educational systems, regional universities, or Arabic-language curricula are represented. Dataset analysis confirms the moral_scenarios subject explicitly anchors ground truth to 'ordinary moral standards in the US as of 2020' (D12–D14) — a direct cultural-frame conflict for a deployment serving users in Arab countries. Web evidence (CAMEL, MENAValues) empirically confirms LLMs trained on such data exhibit Western/Anglocentric bias in Arab contexts.

ID	Evidence
Q23	"The questions in the dataset were manually collected by graduate and undergraduate students from freely available sources online." (p.3)
Q24	"These include practice questions for tests such as the Graduate Record Examination and the United States Medical Licensing Examination." (p.3)
Q25	"It also includes questions designed for undergraduate courses and questions designed for readers of Oxford University Press books." (p.3)
Q83	"We collected approximately 2,000 additional Professional Law training examples." (p.8)
Q85	"We also had RoBERTa continue pretraining on approximately 1.6 million legal case summaries using Harvard's Law Library case law corpus case.law, but after fine-tuning it only attained 36.1% accuracy." (p.8)
D12	For which of these two scenarios does the main character do something clearly morally wrong, according to ordinary moral standards in the US as of 2020? — Explicitly frames moral judgment via US 2020 standards
D13	according to ordinary moral standards in the US as of 2020? Scenario 1: When I went through the road block I gave the policeman my brothers ID. — US-anchored moral framing
D14	according to ordinary moral standards in the US as of 2020? Scenario 2: I was due to start work soon, so I went to a baseball game. — US cultural reference (baseball) embedded in moral reasoning

Strengths

- Source materials are documented (GRE, USMLE, Oxford Press) [\[Q24, Q25\]](#), enabling provenance audit
- Some non-Western primary sources do appear in world_history (Nkrumah, Sasanian inscription) showing the subject is not exclusively Western European [\[D16, D37\]](#)
- Contamination-mitigation steps documented (questions/answers on separate pages) [\[Q112\]](#) reduce one form of construct-irrelevant variance

ID	Evidence
Q24	"These include practice questions for tests such as the Graduate Record Examination and the United States Medical Licensing Examination." (p.3)
Q25	"It also includes questions designed for undergraduate courses and questions designed for readers of Oxford University Press books." (p.3)
Q112	"We also note that most of our questions came from PDFs or websites where questions and answers are on separate pages." (p.14)

D16	Excerpt from the inscription of Kerdir at Naqsh-e-Rustam, Persia, late third century C.E. ... Jews, Buddhists, Hindus, Nasoreans, Christians, Maktak, and Manichaeans in the empire were smitten — Near Eastern (Persian) source — the only example touching the broader Middle East region
D37	Kwame Nkrumah, Neo-Colonialism, 1965 ... 'A State in the grip of neo-colonialism is not master of its own destiny.' — African post-colonial context — non-Western source, though not Arab

Checklist

Checklist item definitions:

ID	Assessment Question
IC-1	Do inputs require region-specific cultural/geographic/dialectal knowledge?
IC-2	Does culturally sensitive content align with target culture?
IC-3	Flag inputs assuming Western-specific knowledge
IC-4	Regional annotator recruitment for culturally sensitive instances
IC-5	Document content issues harming content validity

Checklist responses:

IC-1: Yes — the deployment requires Arab cultural, geographic, and (for the Arabic-learning sub-cohort) dialectal knowledge. MMLU inputs require almost exclusively US/Western cultural and institutional knowledge (D1–D11, D32–D35).

IC-2: Culturally sensitive content (moral_scenarios) is misaligned: explicitly labeled by 'US standards as of 2020' (D12), conflicting with Arab regional norms on alcohol, gender, and family across all eight deployment countries.

IC-3: Multiple inputs require Western-specific knowledge that does not transfer: US tort/contract law [D6, D8], US Social Security (D32), US ethnic demographics (D33), Burgess concentric zone urban model (D28), US contraceptive statistics (D34).

IC-4: INSUFFICIENT DOCUMENTATION — MMLU paper does not document any regional annotator review. ArabicMMLU and ILMAAM provide separate Arab-annotated supplements [WEB-2, WEB-5] but are external to MMLU.

IC-5: Documented: no Arab educational sources [Q24, Q25]; no Arabic-language content; explicit US moral framing in moral_scenarios [D12]; cross-lingual value shifts empirically demonstrated for Arabic vs. English prompts [WEB-11].

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Q85	"We also had RoBERTa continue pretraining on approximately 1.6 million legal case summaries using Harvard's Law Library case law corpus case.law, but after fine-tuning it only attained 36.1% accuracy." (p.8)
Q164	"Krete is an unmarried taxpayer with income exclusively from wages. By December 31, year 1, Krete's employer has withheld \$16,000 in federal income taxes and Krete has made no estimated tax payments. On April 15, year 2, Krete timely filed for an extension request to file her individual tax return, and paid \$300 of additional taxes. Krete's year 1 tax liability was \$16,500 when she timely filed her return on April 30, year 2, and paid the remaining tax liability balance. What amount would be subject to the penalty for underpayment of estimated taxes? (A) \$0 (B) \$500 (C) \$1,650 (D) \$16,500" (p.24)
D1	The right of American citizens to petition the government for a redress of grievances is protected under the First Amendment — US constitutional law — exclusively US institutional framing
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Information Gaps

- Inter-annotator agreement and annotator demographics for MMLU are not documented in the paper
- Specific framing of any 1948 war / Palestinian / Levantine history questions, if present anywhere in MMLU (FV3)

Requires Expert Verification

- Arab regional expert review of any world_religions Islamic content for framing perspective
- Arab expert review of any Middle East / Ottoman / 20th-century Arab references in world_history

Remediation

Gap 1: All MMLU source materials are Anglophone Western institutions; moral_scenarios explicitly encodes US-2020 standards as ground truth.

Recommendation: Supplement with ILMAAM [WEB-5] for culturally aligned content and explicitly exclude moral_scenarios from the deployment evaluation suite. Document the Western-source provenance of any MMLU subjects retained.

ID	Evidence
WEB-5	ILMAAM (2025) identified significant cultural misalignments in religion and morality content; 11 expert annotators reviewed for cultural appropriateness, bias, religious sensitivity (https://aclanthology.org/2025.loreslm-1.29/)

Input Form (IF)

Score: 3/5 — Moderate gaps | Confidence: high

Benchmark: MMLU | Context: Non-Arab Tourists and Expats in Eight Arab Countries

Input Form definition: Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

MMLU's text-only English MCQ format [\[Q3, Q56, Q71\]](#) matches the deployment's primary interaction modality (English text), which the elicitation rates MODERATE priority. However, the Arabic-learning secondary cohort cannot be evaluated by MMLU at all, and MENAValues research [\[WEB-11\]](#) documents cross-lingual value shifts indicating that English MMLU performance does not predict Arabic-prompt behavior. Format brittleness (e.g., separator-token sensitivity [\[Q105\]](#)) adds modest external-validity concern but is not region-specific.

ID	Evidence
Q3	"We design the benchmark to measure knowledge acquired during pretraining by evaluating models exclusively in zero-shot and few-shot settings." (p.1)
Q56	"We feed GPT-3 prompts like that shown in Figure 1a. We begin each prompt with "The following are multiple choice questions (with answers) about [subject]." For zero-shot evaluation, we append the question to the prompt. For few-shot evaluation, we add up to 5 demonstration examples with answers to the prompt before appending the question. All prompts end with "Answer: ". The model then produces probabilities for the tokens "A," "B," "C," and "D," and we treat the highest probability option as the prediction. For consistent evaluation, we create a dev set with 5 fixed few-shot examples for each subject." (p.6)
Q71	"Existing large-scale NLP models, such as GPT-3, do not incorporate multimodal information, so we design our benchmark to capture a diverse array of tasks in a text-only format." (p.7)
Q105	"If we remove the </s> from the input, accuracy declines by several percentage points." (p.12)
WEB-11	MENAValues evaluates LLMs across neutral, personalized, and observer framings — illustrating richer output-form evaluation that MMLU's MCQ accuracy cannot match (https://arxiv.org/html/2510.13154)

Strengths

- English text-only MCQ format aligns with the primary user cohort's interaction modality (Strength 2; D6, D7)
- Zero-shot/few-shot evaluation paradigm [\[Q3\]](#) matches the open-domain generalization required for tourist/expat queries (Strength 4)
- Formal English register is appropriate for educated adult tourist/expat users [\[Q56\]](#)

ID	Evidence
Q3	"We design the benchmark to measure knowledge acquired during pretraining by evaluating models exclusively in zero-shot and few-shot settings." (p.1)
Q56	"We feed GPT-3 prompts like that shown in Figure 1a. We begin each prompt with "The following are multiple choice questions (with answers) about [subject]." For zero-shot evaluation, we append the question to the prompt. For few-shot evaluation, we add up to 5 demonstration examples with answers to the prompt before appending the question. All prompts end with "Answer: ". The model then produces probabilities for the tokens "A," "B," "C," and "D," and we treat the highest probability option as the prediction. For consistent evaluation, we create a dev set with 5 fixed few-shot examples for each subject." (p.6)
D6	A truck driver was assigned to drive a huge tractor-trailer loaded with logs... the girl was struck by the tractor-trailer and was seriously injured — US tort law scenario
D7	A patient filed a medical malpractice action against a hospital in federal court — US federal court procedure

Checklist

Checklist item definitions:

ID	Assessment Question
IF-1	Compare signal distributions between source and target
IF-2	Check if regional infrastructure supports data capture specs
IF-3	Identify domain-specific form differences
IF-4	Document form mismatches harming external validity

Checklist responses:

IF-1: MMLU is exclusively English text [Q71]; primary deployment interaction is English text — match. Arabic-script and dialect inputs from the secondary cohort are unsupported by MMLU and require separate benchmarking [WEB-1, WEB-2].

IF-2: Regional infrastructure not a concern — tourists/expats use standard text-based devices; deployment YAML notes this is not a connectivity gap context.

IF-3: DialectalArabicMMLU [WEB-1] covers Moroccan, Egyptian, Emirati, Saudi dialects (4 of 8 countries) but not Kuwaiti, Jordanian, Palestinian, or Lebanese dialects; format-level dialect sensitivity is unaddressed in MMLU.

IF-4: Form mismatch is moderate: English MCQ matches primary cohort; Arabic-language input form is entirely unevaluated by MMLU. Cross-lingual value shifts [WEB-11] mean MMLU scores cannot be assumed to transfer to Arabic prompts.

ID	Evidence
Q71	“Existing large-scale NLP models, such as GPT-3, do not incorporate multimodal information, so we design our benchmark to capture a diverse array of tasks in a text-only format.” (p.7)
WEB-1	DialectalArabicMMLU covers five Arabic dialects (Syrian, Egyptian, Emirati, Saudi, Moroccan) with substantial cross-dialect performance variation (https://arxiv.org/abs/2510.27543)
WEB-2	ArabicMMLU is the closest available substitute for the Arabic-language secondary cohort (14,575 MSA questions) (https://aclanthology.org/2024.findings-acl.334/)
WEB-11	MENAValues evaluates LLMs across neutral, personalized, and observer framings — illustrating richer output-form evaluation that MMLU’s MCQ accuracy cannot match (https://arxiv.org/html/2510.13154)

Information Gaps

- Tourist/expat device interface preferences (mobile vs. desktop) not searchable at required precision (FV10)
- Whether Arabic-learning users predominantly prompt in MSA vs. dialect

Requires Expert Verification

- Empirical sizing of the Arabic-prompting sub-cohort relative to the English primary cohort

Remediation

Gap 1: English-only format does not evaluate the Arabic-learning sub-cohort.

Recommendation: Add DialectalArabicMMLU [WEB-1] for Moroccan/Egyptian/Emirati/Saudi dialect coverage and ArabicMMLU [WEB-2] for MSA. Note that Kuwaiti, Jordanian, Palestinian, and Lebanese dialects remain unbenchmarked — flag as residual gap.

ID	Evidence
WEB-1	DialectalArabicMMLU covers five Arabic dialects (Syrian, Egyptian, Emirati, Saudi, Moroccan) with substantial cross-dialect performance variation (https://arxiv.org/abs/2510.27543)
WEB-2	ArabicMMLU is the closest available substitute for the Arabic-language secondary cohort (14,575 MSA questions) (https://aclanthology.org/2024.findings-acl.334/)

Output Ontology (OO)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: MMLU | Context: Non-Arab Tourists and Expats in Eight Arab Countries

Output Ontology definition: Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

MMLU's single-answer four-option MCQ schema [\[Q49, Q56\]](#) is structurally incompatible with the deployment's required output behavior — acknowledging multiple perspectives on contested topics (1948 war/Nakba, Western Sahara, Lebanese Civil War, Palestinian statehood, Jerusalem status), per elicitation Q3 and the contested-topics YAML section. Every sampled MMLU example has exactly one labeled correct answer with no mechanism for multi-perspective or uncertainty-flagging responses. The authors note near-random model accuracy on 'morality and law' [\[Q8, Q14\]](#) but do not address the structural taxonomy issue. No identified Arab-focused benchmark uses multi-perspective scoring (FV5). Given OO is HIGH priority, this is a fundamental structural-validity violation.

ID	Evidence
Q3	"We design the benchmark to measure knowledge acquired during pretraining by evaluating models exclusively in zero-shot and few-shot settings." (p.1)
Q8	"Worse, they still have near-random accuracy on some socially important subjects such as morality and law." (p.1)
Q14	"Our results indicate that while recent advances have been impressive, state-of-the-art models still struggle at learning and applying knowledge from pretraining. The tasks with near-random accuracy include calculation-heavy subjects such as physics and mathematics and subjects related to human values such as law and morality." (p.2)
Q49	"To measure performance on our multitask test, we compute the classification accuracy across all examples and tasks." (p.5)
Q56	"We feed GPT-3 prompts like that shown in Figure 1a. We begin each prompt with "The following are multiple choice questions (with answers) about [subject]." For zero-shot evaluation, we append the question to the prompt. For few-shot evaluation, we add up to 5 demonstration examples with answers to the prompt before appending the question. All prompts end with "Answer: ". The model then produces probabilities for the tokens "A," "B," "C," and "D," and we treat the highest probability option as the prediction. For consistent evaluation, we create a dev set with 5 fixed few-shot examples for each subject." (p.6)

Strengths

- Difficulty stratification (Elementary/High School/College/Professional) [\[Q26\]](#) does provide useful granularity within the MCQ schema
- Aggregation across four broad clusters (humanities, social sciences, STEM, other) [\[Q99-Q101\]](#) supports diagnostic interpretation

ID	Evidence
Q26	"Some tasks cover a subject, like psychology, but at a specific level of difficulty, such as "Elementary," "High School," "College," or "Professional."" (p.3)
Q99	"RoBERTa-base attains an overall accuracy of 27.9%, with 27.9% accuracy for the humanities, 28.8% for social sciences, 27.0% for STEM, and 27.7% for other." (p.12)
Q100	"ALBERT-xxlarge attains an accuracy of 27.1%, with 27.2% accuracy for the humanities, 25.7% for the social sciences, 27.7% for STEM, and 27.9% for other." (p.12)
Q101	"GPT-2 attains an accuracy of 32.4%, with 32.8% accuracy for the humanities, 33.3% for the social sciences, 30.2% for STEM, and 33.1% for other." (p.12)

Checklist

Checklist item definitions:

ID	Assessment Question
OO-1	Review output label categories for regional relevance
OO-2	Identify missing regionally specific categories
OO-3	Flag categories encoding non-regional values
OO-4	Consider stakeholder-driven taxonomy redesign
OO-5	Document taxonomy issues harming structural/content validity

Checklist responses:

OO-1: MCQ A/B/C/D label categories are regionally neutral in form but cannot encode multi-perspective or contested-topic acknowledgment, which the deployment explicitly requires per elicitation Q3.

OO-2: Missing categories: no schema slot for 'this is contested,' 'depends on country,' or 'multiple perspectives apply' — required for Palestinian narrative, Western Sahara, Lebanese Civil War, etc. [contested_topics YAML section].

OO-3: moral_scenarios schema explicitly encodes 'US standards as of 2020' as the decision rule (D12), a non-regional value framing.

OO-4: Stakeholder-driven redesign would require moving beyond MCQ entirely; no Arab-focused benchmark using multi-label or open-ended contested-history scoring has been identified (FV5).

OO-5: Documented structural-validity violation: the construct (multi-perspective handling of contested Arab regional topics) cannot be represented in the MMLU output schema; MMLU paper itself flags weak performance on 'human values such as law and morality' [Q14, Q15] without addressing the schema issue.

ID	Evidence
Q3	"We design the benchmark to measure knowledge acquired during pretraining by evaluating models exclusively in zero-shot and few-shot settings." (p.1)
Q8	"Worse, they still have near-random accuracy on some socially important subjects such as morality and law." (p.1)
Q14	"Our results indicate that while recent advances have been impressive, state-of-the-art models still struggle at learning and applying knowledge from pretraining. The tasks with near-random accuracy include calculation-heavy subjects such as physics and mathematics and subjects related to human values such as law and morality." (p.2)
Q15	"This second weakness is particularly concerning because it will be important for future models to have a strong understanding of what is legal and what is ethical." (p.2)
Q49	"To measure performance on our multitask test, we compute the classification accuracy across all examples and tasks." (p.5)
Q56	"We feed GPT-3 prompts like that shown in Figure 1a. We begin each prompt with 'The following are multiple choice questions (with answers) about [subject].' For zero-shot evaluation, we append the question to the prompt. For few-shot evaluation, we add up to 5 demonstration examples with answers to the prompt before appending the question. All prompts end with 'Answer: '. The model then produces probabilities for the tokens 'A,' 'B,' 'C,' and 'D,' and we treat the highest probability option as the prediction. For consistent evaluation, we create a dev set with 5 fixed few-shot examples for each subject." (p.6)
D12	For which of these two scenarios does the main character do something clearly morally wrong, according to ordinary moral standards in the US as of 2020? — Explicitly frames moral judgment via US 2020 standards
WEB-5	ILMAAM (2025) identified significant cultural misalignments in religion and morality content; 11 expert annotators reviewed for cultural appropriateness, bias, religious sensitivity (https://aclanthology.org/2025.loreslm-1.29/)
WEB-11	MENAValues evaluates LLMs across neutral, personalized, and observer framings — illustrating richer output-form evaluation that MMLU's MCQ accuracy cannot match (https://arxiv.org/html/2510.13154)

Information Gaps

- Whether any planned MMLU successor or supplement supports multi-perspective scoring (none identified per FV5)

Requires Expert Verification

- Region-stakeholder design of an output schema that accommodates contested-topic acknowledgment

Remediation

Gap 1: Single-answer MCQ schema cannot represent multi-perspective handling of contested topics (1948 war, Western Sahara, Lebanese Civil War, Palestinian statehood, Jerusalem) — the deployment's explicitly required output behavior per elicitation Q3.

Recommendation: Develop a custom evaluation harness for contested topics using multi-perspective rubrics with Arab regional expert adjudication; do not rely on MMLU MCQ accuracy for these topics. MENAValues [\[WEB-11\]](#) cross-lingual framing methodology is the closest published model.

ID	Evidence
Q3	"We design the benchmark to measure knowledge acquired during pretraining by evaluating models exclusively in zero-shot and few-shot settings." (p.1)
WEB-11	MENAValues evaluates LLMs across neutral, personalized, and observer framings — illustrating richer output-form evaluation that MMLU's MCQ accuracy cannot match (https://arxiv.org/html/2510.13154)

Output Content (OC)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: MMLU | Context: Non-Arab Tourists and Expats in Eight Arab Countries

Output Content definition: Whether ground-truth labels align with the judgments of regional stakeholders.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

MMLU annotator population is graduate/undergraduate students from US institutions [Q23] with Mechanical Turk and US 95th-percentile test-taker baselines [Q31, Q33, Q34] — no Arab regional stakeholders. Ground-truth labels reflect Western academic framing, with moral_scenarios explicitly encoding 'US standards as of 2020' (D12). Inter-annotator agreement is not documented. Empirical evidence (CAMEL, ILMAAM) confirms Western framing of Arab content in LLM evaluation systematically diverges from Arab stakeholder judgments [WEB-5, WEB-15]. With OC rated HIGH and the deployment serving eight Arab countries with contested historical narratives, this is a major convergent- and external-validity violation.

ID	Evidence
Q23	"The questions in the dataset were manually collected by graduate and undergraduate students from freely available sources online." (p.3)
Q31	"Unspecialized humans from Amazon Mechanical Turk obtain 34.5% accuracy on this test." (p.3)
Q33	"For example, real-world test-taker human accuracy at the 95th percentile is around 87% for US Medical Licensing Examinations, and these questions make up our "Professional Medicine" task." (p.3)
Q34	"If we take the 95th percentile human test-taker accuracy for exams that build up our test, and if we make an educated guess when such information is unavailable, we then estimate that expert-level accuracy is approximately 89.8%." (p.3)
D12	For which of these two scenarios does the main character do something clearly morally wrong, according to ordinary moral standards in the US as of 2020? — Explicitly frames moral judgment via US 2020 standards
WEB-5	ILMAAM (2025) identified significant cultural misalignments in religion and morality content; 11 expert annotators reviewed for cultural appropriateness, bias, religious sensitivity (https://aclanthology.org/2025.loreslm-1.29/)
WEB-15	Naous et al. 2024 (CAMEL) empirically demonstrate LLM Western cultural bias in Arabic prompts (https://venturebeat.com/ai/large-language-models-exhibit-significant-western-cultural-bias-study-finds)

Strengths

- Annotator source institutions are documented [Q23, Q24, Q25], enabling diagnosis even though the demographics themselves are problematic
- Domain-expert performance is benchmarked against US 95th-percentile test-takers [Q33], providing a calibrated US-centric ceiling — useful as a baseline even if not regionally valid

ID	Evidence
Q23	"The questions in the dataset were manually collected by graduate and undergraduate students from freely available sources online." (p.3)
Q24	"These include practice questions for tests such as the Graduate Record Examination and the United States Medical Licensing Examination." (p.3)
Q25	"It also includes questions designed for undergraduate courses and questions designed for readers of Oxford University Press books." (p.3)
Q33	"For example, real-world test-taker human accuracy at the 95th percentile is around 87% for US Medical Licensing Examinations, and these questions make up our "Professional Medicine" task." (p.3)

Checklist

Checklist item definitions:

ID	Assessment Question
OC-1	Do ground-truth labels reflect regional stakeholder perspectives?
OC-2	Assess potential annotator-regional population disagreement
OC-3	Review annotator demographics from documentation
OC-4	Consider label re-annotation by regional annotators
OC-5	Review aggregation methods for minority perspective erasure
OC-6	Document label issues harming convergent/external validity

Checklist responses:

OC-1: Ground-truth labels do not reflect Arab regional stakeholder perspectives. moral_scenarios labels are explicitly US-2020-anchored (D12); professional_law labels reflect US common law (D6–D8); us_foreign_policy labels reflect US institutional framing (D9–D11).

OC-2: High disagreement expected on contested topics (Palestinian narrative, Western Sahara, Jerusalem) where Arab stakeholder judgments diverge from US-academic-sourced labels [\[contested_topics YAML, WEB-5\]](#).

OC-3: MMLU paper does not document annotator demographics beyond 'graduate and undergraduate students' [\[Q23\]](#) — INSUFFICIENT DOCUMENTATION on country, ethnicity, or expertise composition.

OC-4: Re-annotation is what ArabicMMLU [\[WEB-2, WEB-6\]](#) and ILMAAM [\[WEB-5\]](#) effectively do — but for separate Arabic-content benchmarks, not for MMLU itself.

OC-5: INSUFFICIENT DOCUMENTATION — MMLU paper does not describe label aggregation methods or treatment of annotator disagreement; whether minority perspectives are erased cannot be assessed from the paper.

OC-6: Confirmed convergent- and external-validity violations: US-centric annotation [\[Q23, Q31, Q33\]](#), explicit US-2020 moral framing (D12), no Arab expert review documented; CAMEL evidence of LLM Western bias [\[WEB-15\]](#) confirms practical impact.

ID	Evidence
Q23	"The questions in the dataset were manually collected by graduate and undergraduate students from freely available sources online." (p.3)
Q31	"Unspecialized humans from Amazon Mechanical Turk obtain 34.5% accuracy on this test." (p.3)
Q33	"For example, real-world test-taker human accuracy at the 95th percentile is around 87% for US Medical Licensing Examinations, and these questions make up our "Professional Medicine" task." (p.3)
Q79	"They are notably poor at modeling human (dis)approval, as evident by the low performance on the Professional Law and Moral Scenarios tasks." (p.8)
D6	A truck driver was assigned to drive a huge tractor-trailer loaded with logs... the girl was struck by the tractor-trailer and was seriously injured — US tort law scenario
D7	A patient filed a medical malpractice action against a hospital in federal court — US federal court procedure
D8	A buyer filed a lawsuit against a seller based on a written contract allegedly executed at the time of the sale of the seller's hot dog stand — US contract law
D9	Why is NSC 68 seen as a turning point in US Cold War Foreign policy? ... It indicated a shift towards military containment — US Cold War policy — framed entirely from US perspective
D10	In American government, the power to declare war rests with Congress. — US constitutional structure
D11	Revelations that the NSA was monitoring the communications of American citizens without obtaining warrants — US domestic surveillance scandal
D12	For which of these two scenarios does the main character do something clearly morally wrong, according to ordinary moral standards in the US as of 2020? — Explicitly frames moral judgment via US 2020 standards
WEB-2	ArabicMMLU is the closest available substitute for the Arabic-language secondary cohort (14,575 MSA questions) (https://aclanthology.org/2024.findings-acl.334/)
WEB-5	ILMAAM (2025) identified significant cultural misalignments in religion and morality content; 11 expert annotators reviewed for cultural appropriateness, bias, religious sensitivity (https://aclanthology.org/2025.loreslm-1.29/)

WEB-6	ArabicMMLU annotated by 10 Arab native speakers from Jordan, Egypt, Lebanon, UAE, KSA — illustrating the regional annotation MMLU lacks (https://github.com/mbzuai-nlp/ArabicMMLU)
WEB-15	Naous et al. 2024 (CAMEL) empirically demonstrate LLM Western cultural bias in Arabic prompts (https://venturebeat.com/ai/large-language-models-exhibit-significant-western-cultural-bias-study-finds)

Information Gaps

- Annotator country/ethnicity composition beyond 'students'
- Inter-annotator agreement statistics (not reported in MMLU paper)
- Aggregation methodology for any items where annotators disagreed (not reported)

Requires Expert Verification

- Arab regional expert relabeling of any Arab-region or Islamic-content questions in MMLU (e.g., world_religions Islamic items, world_history Middle East items)

Remediation

Gap 1: Ground-truth labels were determined by US graduate/undergraduate annotators with no documented Arab regional stakeholder review.

Recommendation: For any MMLU items retained that touch Arab or Islamic content (world_religions Islamic items, world_history Middle East items, geography items), commission Arab regional expert relabeling. Use ILMAAM's 11-expert annotation methodology [WEB-5] as a template.

ID	Evidence
WEB-5	ILMAAM (2025) identified significant cultural misalignments in religion and morality content; 11 expert annotators reviewed for cultural appropriateness, bias, religious sensitivity (https://aclanthology.org/2025.loreslm-1.29/)

Output Form (OF)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: MMLU | Context: Non-Arab Tourists and Expats in Eight Arab Countries

Output Form definition: Whether the expected output modality matches regional deployment needs and accessibility.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Both MMLU and the deployment use text output, satisfying basic modality match. However, MMLU's classification-accuracy metric over A/B/C/D tokens [Q49, Q56] cannot evaluate the open-ended, multi-perspective explanatory responses the deployment's tourist/expat assistant produces. Calibration evaluation [Q64–Q67] is a useful auxiliary signal — GPT-3's 24% confidence-accuracy gap [Q66] is informative — but does not address the form mismatch. With OF rated MODERATE, this is a real but partial concern: MMLU can score factual recall in MCQ form but not the explanatory output form the deployment expects.

ID	Evidence
Q49	"To measure performance on our multitask test, we compute the classification accuracy across all examples and tasks." (p.5)
Q56	"We feed GPT-3 prompts like that shown in Figure 1a. We begin each prompt with "The following are multiple choice questions (with answers) about [subject]." For zero-shot evaluation, we append the question to the prompt. For few-shot evaluation, we add up to 5 demonstration examples with answers to the prompt before appending the question. All prompts end with "Answer: ". The model then produces probabilities for the tokens "A," "B," "C," and "D," and we treat the highest probability option as the prediction. For consistent evaluation, we create a dev set with 5 fixed few-shot examples for each subject." (p.6)
Q64	"We should not trust a model's prediction unless the model is calibrated, meaning that its confidence is a good estimate of the actual probability the prediction is correct." (p.7)
Q65	"We evaluate the calibration of GPT-3 by testing how well its average confidence estimates its actual accuracy for each subject." (p.7)
Q66	"Its confidence is only weakly related to its actual accuracy in the zero-shot setting, with the difference between its accuracy and confidence reaching up to 24% for some subjects." (p.7)
Q67	"Another calibration measure is the Root Mean Squared (RMS) calibration error (Hendrycks et al., 2019a; Kumar et al., 2019)." (p.7)

Strengths

- Text output modality matches deployment text interface
- Calibration metrics (RMS calibration error, confidence-accuracy correlation) [Q65, Q67] provide useful auxiliary signals on whether models 'know what they don't know' [Q16] — directly relevant to a tourist/expat assistant that should hedge on contested topics
- Zero-shot/few-shot evaluation regime [Q3] aligns with deployment's open-domain generalization requirement (Strength 4)

ID	Evidence
Q3	"We design the benchmark to measure knowledge acquired during pretraining by evaluating models exclusively in zero-shot and few-shot settings." (p.1)
Q16	"Worryingly, we also find that GPT-3 does not have an accurate sense of what it does or does not know since its average confidence can be up to 24% off from its actual accuracy." (p.2)
Q65	"We evaluate the calibration of GPT-3 by testing how well its average confidence estimates its actual accuracy for each subject." (p.7)
Q67	"Another calibration measure is the Root Mean Squared (RMS) calibration error (Hendrycks et al., 2019a; Kumar et al., 2019)." (p.7)

Checklist

Checklist item definitions:

ID	Assessment Question
OF-1	Does output modality match regional deployment needs?
OF-2	Assess text-to-speech availability for speech output
OF-3	Consider literacy rates and accessibility requirements
OF-4	Document form mismatches harming external validity

Checklist responses:

OF-1: Modality matches (text), but output granularity mismatches: MCQ token probability [Q56] vs. open-ended explanation expected by deployment users.

OF-2: Not applicable — text-based deployment, no TTS requirement documented.

OF-3: Tourist/expat target population is literate adult readers; literacy not a concern. INSUFFICIENT DOCUMENTATION on accessibility-specific output requirements.

OF-4: Form mismatch documented: MCQ scoring [Q49] cannot credit appropriately-hedged multi-perspective open-ended responses; the authors note their evaluation format 'is not identical to the format in which information is acquired during pretraining' [Q76].

ID	Evidence
Q49	"To measure performance on our multitask test, we compute the classification accuracy across all examples and tasks." (p.5)
Q56	"We feed GPT-3 prompts like that shown in Figure 1a. We begin each prompt with "The following are multiple choice questions (with answers) about [subject]." For zero-shot evaluation, we append the question to the prompt. For few-shot evaluation, we add up to 5 demonstration examples with answers to the prompt before appending the question. All prompts end with "Answer: ". The model then produces probabilities for the tokens "A," "B," "C," and "D," and we treat the highest probability option as the prediction. For consistent evaluation, we create a dev set with 5 fixed few-shot examples for each subject." (p.6)
Q76	"Importantly, the format of our evaluation is not identical to the format in which information is acquired during pretraining." (p.8)
WEB-11	MENAValues evaluates LLMs across neutral, personalized, and observer framings — illustrating richer output-form evaluation that MMLU's MCQ accuracy cannot match (https://arxiv.org/html/2510.13154)

Information Gaps

- Whether the deployed system generates free-form text or also offers MCQ-style confirmations (interface design not specified in YAML)

Requires Expert Verification

- Whether deployment evaluation will supplement MMLU MCQ accuracy with open-ended generation metrics (e.g., MENAValues-style prompting)

Remediation

Gap 1: MCQ accuracy metric cannot evaluate open-ended explanatory or perspective-acknowledging responses that the deployed assistant produces.

Recommendation: Add open-ended generation evaluation with rubric-based scoring (cultural appropriateness, multi-perspective acknowledgment, calibrated hedging on contested topics). Re-evaluate every 6–12 months given rapid Arabic-benchmark ecosystem growth [WEB-12].

ID	Evidence
WEB-12	Survey of Arabic benchmarks notes gaps in coverage of North Africa, Levant, Iraq even within the Arabic NLP ecosystem (https://arxiv.org/html/2510.13430v1)

